ST-Net: Scattering Topology Network for Aircraft Classification in High-Resolution SAR Images

Yuzhuo Kang, Zhirui Wang, Haoyu Zuo, Yidan Zhang, Zhujun Yang, Xian Sun, *Senior Member, IEEE*, and Kun Fu

***Abstract*— Aircraft classification in synthetic aperture radar (SAR) images plays a considerable role in global region management and surveillance. Recently, deep learning has been applied to solve the classification problem and made significant progress. Due to the imaging variability at different angles and component scattering discreteness in SAR images, previous works have had difficulty in achieving desirable classification results. To address these issues, we study the positional and semantic relationship between the scattering points and propose an innovative scattering topology network (ST-Net) in this article. First, considering the diversity of imaging results caused by different target attitude angles, we extract and transform the scattering cluster centers to update the information of various categories. It can guide the model to strengthen the discriminative features and mitigate the impact of imaging variability on classification performance. Second, a novel scattering topology module (STM) is introduced to model the spatial relationships and semantic information interaction of discrete scattering points. In this process, the topology relations and scattering characteristics are enhanced for further accurate classification. Third, context attention excitation (CAE) is designed to capture significant global and semantic information, which is conducive to suppressing background interference and reducing category confusion. In conclusion, the ST-Net is presented with the SAR imaging mechanism and the topology geometric representation of aircraft. We construct the SAR aircraft category dataset (SAR-ACD) and conduct extensive experiments on it to show the effectiveness of ST-Net, which illustrates that our method achieves superior classification performance.**

***Index Terms*— Aircraft classification, context attention excitation (CAE), scattering point, synthetic aperture radar (SAR), topology structure.**

I. Introduction

SYNTHETIC aperture radar (SAR) is an active microwave imaging system and can observe the Earth in all-time and all-weather conditions, which is not limited by natural conditions such as weather and light. This advantage makes it widely used in many fields [1], [2], [3], [4], [5], [6], [7], [8], such as object detection and disaster monitoring. With the rapid development of SAR imaging techniques, SAR target classification has drawn a lot of attention and is continuously developed. Among different targets, aircraft classification plays a critical role in global region management and surveillance.

The process of SAR target classification consists of feature extraction and classifier, in which feature extraction is the foundation for distinguishing different categories. In traditional methods, the typical feature representations for classification include geometric structure features, texture features, and grayscale statistical features. Combined with the geometric features of the aircraft, Chen et al. [9] extracted the salient point vector of the targets and matched it with the template containing two relaxation variables. Considering the plentiful texture features, Huan and Yang [10] utilized the Gabor wavelet transform to obtain the texture information from the SAR images. Furthermore, Fu et al. [11] used statistical methods to model the scattering feature of aircraft and then proposed a sample decision optimization algorithm to achieve the target recognition. In general, the traditional methods usually have excellent interpretability for features, but they rely on hand-designed parameters and rules, which limits their feature description capabilities.

Recently, driven by the powerful feature extraction ability of deep learning methods, convolutional neural networks (CNNs) have been introduced into the SAR domain [12], [13], [14] and made remarkable breakthroughs in terms of the classification problem. Considering the problem of data acquisition, Malmgren-Hansen et al. [13] used a computer-aided design (CAD) model to simulate SAR images and initialized the CNN with argument data. To improve the classification accuracy, Wang et al. [14] designed a discrimination framework containing the density and gradient information of SAR images. These methods based on deep learning extract more representative features from the input SAR images and usually achieve good results. However, these methods directly transfer the methods of optical images to SAR images and ignore the connection to SAR aircraft imaging characteristics, resulting in poor generalization ability in complex scenes. Unlike optical images, the targets in SAR images are often composed of scattering points. Effectively combining the information of scattering points is beneficial to improve the performance of classification. Sun et al. [15], [16] extract the strong scattering

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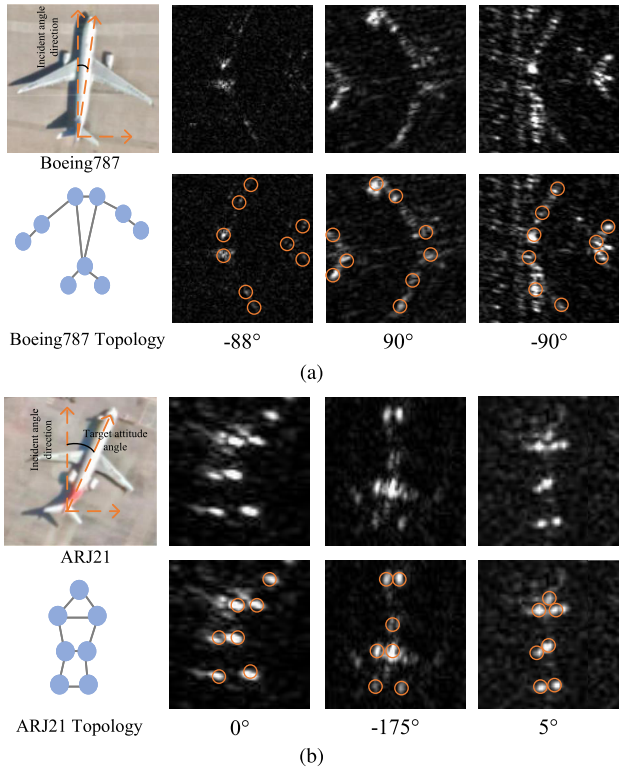


Fig. 1. SAR images of Boeing787 and ARJ21 categories with different target attitude angles and corresponding strong scattering points. The direction of the incident angle is defined as 0°. The target attitude angle is defined from 0° to 180° in the clockwise and 0°−−180° in the counterclockwise direction. (a) Boeing787. (b) ARJ21. (Top) (a) and (b) Optical remote sensing images and SAR images. (Bottom) (a) and (b) Clusters of scattering points, which reflect the topology of the aircraft.

points of targets to identify targets accurately, but they only take the key points as a kind of supervision information and do not make full use of the location and spatial information of scattering points.

In general, the above methods only pay attention to the whole of the targets and abandon the independent component characteristics. Inspired by the deformable part model (DPM) method [17], objects of the same class always have similar parts. Focusing on the aircraft targets in Fig. 1, the texture features of the aircraft are not exactly the same due to the differences in imaging conditions, while they both contain salient discrete structures with consistent distribution. In high-resolution SAR images, the crucial components with dihedral and trihedral parts in the aircraft present as discrete clusters of strong scattering points. These scattering point clusters with the large pixel value of the local area have certain significance and stability, and they contain a large amount of scattering information [9], [18]. In addition, the positional connections of these clusters constitute the unique distribution of the aircraft in each category, which contributes to improving classification accuracy. Consequently, we can introduce the concepts of topology and component distribution into the SAR aircraft model. The effective extraction and utilization of the structural features of the scattering points are beneficial to the fine-grained classification of SAR images.

At present, the methods of structure topology have been studied and extensively applied to object detection [19] and

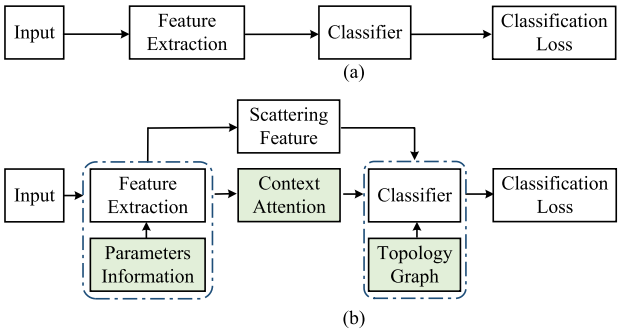


Fig. 2. Classification algorithms of the previous work and ST-Net. (a) Previous general methods for classification. (b) ST-Net (Ours) constructs the topology relationships and combines the scattering characteristics.

classification [20]. These methods usually transform topology data into simple feature representations through some data processing methods, but they ignore the spatial dependence of information on some nodes. In terms of SAR imagery, He et al. [21] detected the nose and tail of the aircraft separately and then leveraged the topology relationship between them to optimize and combine the final results. However, it requires additional component annotations and lacks global information to suppress background interference.

After the above analysis, there exist the following problems in aircraft classification in SAR images. First, due to the SAR imaging mechanism, different condition parameters cause the targets to appear with diverse scattering characteristics. As shown in Fig. 1, the changes in the target attitude angles result in the variability of imaging results for the same category, which brings a notable challenge to the accuracy of aircraft classification. Second, different from the optical images, the prime components of SAR aircraft exhibit discrete strong scattering points. The lack of scattering characteristics and spatial relationships of components increases the difficulty of distinguishing the different categories. Third, owing to the constraints of the receptive field, the global context information is neglected. Coupled with the interference of background clutter in images, it easily leads to prominent category confusion. Hence, the above aspects indicate that SAR aircraft classification is a challenging task.

To address these issues, a novel method named scattering topology network (ST-Net) is presented in this article for aircraft classification. Different from previous methods, ST-Net constructs the graph-based topology relationships and integrates the scattering characteristic of components, as illustrated in Fig. 2. First, ST-Net extracts the scattering cluster centers and corresponding scattering features for each target automatically. The imaging parameters are fused to match and update the scattering feature of different categories, which is conducive to highlighting the differences of strong scattering components on various types of targets. Second, the proposed method takes advantage of the scattering characteristic and topology. We define the cluster centers as nodes and model spatial relations between different nodes as edges. By using the graph convolutional network (GCN), the scattering information and spatial relations are converted to discriminative object representation. Third, equipped with the ability to enhance the

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connection between different channels, the proposed method explores the context information and strengthens the global features to mitigate the effects of background clutter. The overall classification accuracy is significantly improved.

The main contributions are summarized as follows.

1) A novel framework of aircraft classification named ST-Net is proposed in this article. It utilizes the scattering characteristics of aircraft and integrates the spatial topology of scattering cluster centers in the targets. Furthermore, we construct a public SAR aircraft category dataset (SAR-ACD), and the experiments on it demonstrate the superiority and robustness of ST-Net.
2) ST-Net employs an effective scattering-aware feature matching (SAFM) module to construct the scattering feature templates of various categories. It can guide the model to strengthen the discriminative features and alleviate the influence of imaging variability on classification performance.
3) The innovative scattering topology module (STM) based on the GCN is introduced to model the spatial relations and semantic information interaction of the various components at different positions, which is beneficial to enhance the geometric representation of SAR targets.
4) Taking the global information into account, the context attention excitation (CAE) is designed to capture significant integral information about the objects while suppressing background interference. It can reduce category confusion in an effective way.

The results of ablation experiments demonstrate the effectiveness of the ST-Net. The rest of this article is organized as follows. In Section II, we briefly introduce the related work. Section III explains the details of ST-Net for aircraft recognition. In Section IV, the experiment results on SAR-ACD are presented and analyzed. Finally, Section V concludes this article.

II. Related Work

This section first summarizes the methods of SAR target classification. Considering the topology graph structure of the aircraft components, we next introduce the different graph neural networks (GNNs) in Section II-B.

A. SAR Target Classification

The traditional SAR target classification methods first extract some feature vectors of targets and then compare them with the database to get the classification results. They contain two types of approaches: 1) template-based matching [22], [23] and 2) model-based [24], [25] methods. The template-based methods extract some features of the targets and make similarity matching with the template library to determine the target category. Hu et al. [22] adopted the geometric feature, statistical feature, and Hu moment invariant feature fusion decision to classify aircraft targets in SAR images. Furthermore, Pan et al. [23] classified targets by calculating the matching cost function of the peak point set from the target image and template image. This kind of method brings convenience and proficiency to the operation, but it requires extensive template libraries and is inefficient.

In order to reduce storage and improve computing efficiency, the model-based methods [24], [25] are proposed to carry out real-time retrieval and matching. Zhang et al. [24] focused on the CAD model and constructed the circular SAR of the targets by using the principle of catapulting line, which finally completed the classification through online real-time simulation of target SAR images. Besides, Qin et al. [25] utilized the scattering center model to generate simulation images in real-time and matched the features extracted from the images to be recognized with the simulation images. These traditional algorithms have achieved good classification results and have been widely used. However, the modeling process is complex, and it has a high dependence on parameters and low stability.

In recent years, with the development of deep learning, many experts [26], [27], [28], [29], and [30] have paid attention to CNN and applied it to SAR target classification. Chen et al. [26] presented the A-Convnets algorithm to replace the full connection with the convolution layer of sparse connection, which aims to mitigate the growth of the number of free parameters and reduce the risk of overfitting. Wagner [27] utilized the CNN for feature extraction and integrated the gradient information for further classification. In view of the lack of data, strategies such as data augmentation [28], transferring learning [29], and meta-learning [30] have been adopted to address the problem of small sample target classification. Compared with the traditional methods of manually designing features, the CNN-based methods reduce the parameters and provide a strong feature representation ability. However, they lack the combination with the SAR target characteristics and have poor robustness of feature representation.

Different from the above methods, our work combines the SAR imaging principle and models the depth feature vectors of target scattering key points. It increases the distinction of features between classes by learning the scattering templates adaptively from dynamic environment data.

B. Graph Neural Network

The GNN [31] can utilize neural networks to extract features and patterns in graph structure data under non-Euclidean space. Inspired by it, the GCN [32] is proposed to integrate feature and label information of the central and neighbor nodes through the ingenious transformation of convolution operators. In order to explore the connections between different edges, graph attention network (GAT) [33] leverages the attention mechanism [34] to assign different weights to various neighbor nodes. Graph-SAGE [35] increases scalability by learning an aggregator function to implement node embedding inductively.

Recently, driven by the success and excellent performance of GNN on graph-structured data, it has been applied to various tasks in computer vision [36], [37], [38], [39], [40], [41], [42], [43], such as, scene recognition, multilabel classification, and few-shot classification problem. In terms of scene recognition, Peng et al. [36] designed a multitask graph structure for predicting the component objects and

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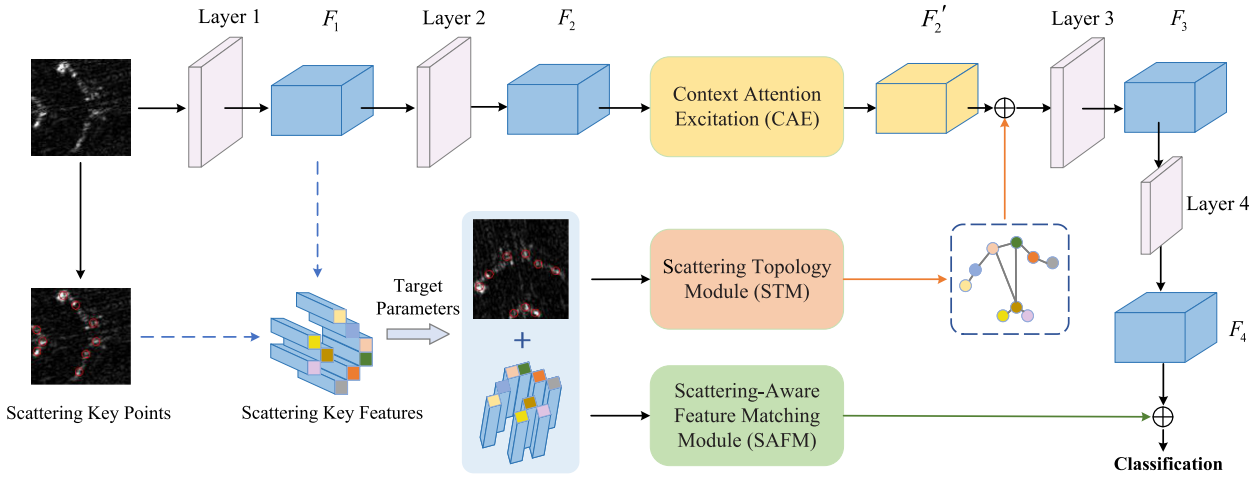


Fig. 3. Overall architecture of ST-Net. First, we extract the scattering key points of the objects and utilize their positions to obtain the corresponding scattering key features. Then, the coordinates converted by target parameters and the corresponding features are sent into the SAFM and STM for constructing the topology relations. In addition, the CAE is added to the backbone to capture more global information adaptively. Finally, the aggregated features are fed into the classification branch to predict the class scores.

facility recognition simultaneously. Chen et al. [38] proposed a novel prototype-agnostic scene recognition that is constructed by the locations and sizes of important regions. For the multilabel classification, considering the attributes of target co-occurrence and label dependencies, Chen et al. [39] model the relations between different target categories in the same images. Wang et al. [40] leveraged the prior knowledge to supplement the statistical label graph, which can take full advantage of label embedding correlations. For the few-shot classification problem, Garcia and Bruna [41] constructed the relationship between the collection of input images and performed the inference on the observed graphical model. Based on the roughly annotated data, Liu et al. [42] proposed the prototype propagation network together with the category graph.

The above methods combined with GNN have achieved good results, which further demonstrates its satisfactory performance in computer vision. Inspired by these studies, we leverage the graph structure to design a topology relation module between different scattering key points, which can enhance the spatial and semantic relationship simultaneously. Besides, the proposed method adopts an unsupervised approach for components, which reduces the demand for manual annotations. The method of exploring the potential of graph structures is beneficial for fine-grained classification.

III. Methodology

In this section, the whole network structure is first introduced in detail. Then, the SAFM devised to highlight the difference in scattering characteristics of various categories is introduced in Section III-B. In addition, to aggregate the discrete components of aircraft in SAR images, we explain a novel STM with the purpose of modeling the positional correlation between strong scattering points. Finally, the CAE module is designed to guide global information adaptation and suppress interference.

A. Proposed Architecture

The overall classification framework of the ST-Net is presented in Fig. 3, which contains four main parts, i.e., a backbone, a feature scattering-aware matching module named SAFM, a feature topology relation network named STM, and a global information enhancement module named CAE.

The deep CNN improves the feature extraction ability in the task of target classification. Based on the generic ResNet-34 [44] architecture, our method is designed with the scattering characteristics analysis, which aims to enhance the topology relations of scattering key points and increase classification accuracy.

Specifically, we first extract the scattering key points of the images and utilize their positions to obtain the corresponding scattering features. Considering the target parameters, i.e., target attitude angles, resolutions, and so on, the coordinates and scattering features can be further transformed to the standard condition. Then the features of neighbor positions are clustered together according to the coordinates, and the scattering component templates of various categories are updated.

In the feature relation network, the STM based on graph structure is adopted to aggregate the discrete components of aircraft. It can reduce the intraclass topology differences adaptively and enhance the discrimination feature between classes.

To mitigate the impact of background interference on target characteristics, we design the CAE to strengthen global information and reduce misclassification. Simultaneously with feature extraction, the proposed CAE can adjust the sizes of the receptive fields and capture significant context dependence. In summarizing, we propose the classification network ST-Net to achieve fine-grained aircraft classification.

B. Scattering-Aware Feature Matching (SAFM)

Under different imaging conditions, an object will present different scattering characteristics in SAR images. The

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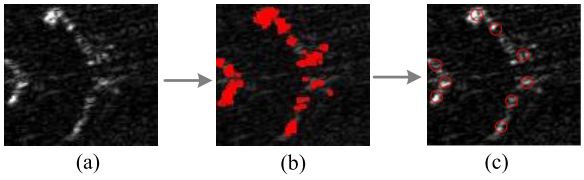


Fig. 4. Illustration of the scattering key points. (a) SAR images of the aircraft. (b) Extracted results by Harris-Laplace detector. (c) Cluster results by K-means algorithm. The red hollow circles are the cluster centers named scattering key points.

changing visual features caused by the various imaging condition increase the difficulty of the classification. However, aircraft in the same category have a similar architectural design, and the number of strong scattering structures such as engines remains stable. Therefore, considering that different classes have exclusive scattering features, we propose a novel grouping and matching model that utilizes the key features to increase the discriminability of different categories. The SAFM contains two parts: extraction and transformation, grouping and matching, which will be introduced in detail.

1) Extraction and Transformation: To obtain the scattering characteristics of the component structure, we first extract the scattering key points from the original clips and obtain the corresponding scattering key features from the feature map. Then the features and coordinates are transformed to the standard condition according to the target parameters.

The strong scattering points have certain significance and stability [15], and they contain a large amount of scattering information [9], [18]. Therefore, we leverage the scattering key points to explore the stable scattering characteristics of the targets. First, the Harris-Laplace detector [45] is performed to extract the important points to reflect the structure outline, which is shown in Fig. 4. Then we utilize the K-means algorithm [46] to divide the extracted points into N clusters with high scattering intensities, which can get rid of the redundant points and obtain a regular structure. According to [47] and [15], the number of cluster centers is set as 9, which can get rid of the redundant points and cover various component structures of different categories as much as possible. We define the N cluster centers as the scattering key points $P = \{P_1, P_2, \ldots, P_N, N = 9\}$. These points represent the local distributions of scattering intensity and structural characteristics of targets, which provides information to enrich the discriminative features of aircraft.

After clustering, the 2-D coordinates $C = \{(x_1, y_1), (x_2, y_2), \ldots, (x_N, y_N)\}$ of the scattering key points on the original images can be obtained. Then we extract the L-neighborhood features around each coordinate from the corresponding position on the feature map $F_1 \in \mathbb{R}^{W_1 \times H_1 \times C_1}$ as the plentiful scattering key features $F_s \in \mathbb{R}^{N \times L \times C_1}$, where W_1 and H_1 represent the width and height of F_1, respectively. Taking the effect of the scattering parameters into account, we combine the target attitude angles and resolutions to carry out physical modeling and transform the coordinates of scattering key points to the standard state.

For the dataset, we supplement the target attitude angles to the annotations in addition to the categories. According to the article [15], the incident angle is defined as 0°. The target attitude angle is defined from 0° to −180° in the counterclockwise direction and 0°–180° in the clockwise direction, as shown in Fig. 1. Furthermore, the different resolutions affect the sizes of the targets, which leads to changes in the distribution of scattering key points. Assuming that the attitude angle of the target and the resolution are θ and ηm in the image, respectively, we can transform the original coordinates to the standard state with 0° and 1 m. This process can be calculated as

$$\tilde{x}_i = (x_i - x_{\text{center}}) \times \cos(\pi/180 \times \theta)/\eta - (y_i - y_{\text{center}}) \times \sin(\pi/180 \times \theta)/\eta + x_{\text{center}}/\eta \quad (1)$$

$$\tilde{y}_i = (x_i - x_{\text{center}}) \times \sin(\pi/180 \times \theta)/\eta + (y_i - y_{\text{center}}) \times \cos(\pi/180 \times \theta)/\eta + y_{\text{center}}/\eta \quad (2)$$

where (x_i, y_i) is the ith scattering key point and $(x_{\text{center}}, y_{\text{center}})$ is the center coordinate of the image. Through this process, we can get the transformed coordinates $\tilde{C} = \{(\tilde{x}_1, \tilde{y}_1), (\tilde{x}_2, \tilde{y}_2), \ldots, (\tilde{x}_N, \tilde{y}_N)\}$ of the scattering key points at 0° attitude angle and 1 m resolution. This conversion is conducive to reducing the influence of different imaging conditions on the distribution of scattering key points.

2) Grouping and Matching: The echo strength of SAR images is often related to the material of the components. Due to the imaging mechanism of SAR, metal materials usually have a large dielectric constant and intense backscattering properties, which can generate strong echo signals. As a result, the nose, engine, fuel tank, and bomb attached to the wingtips emerge the salient scattering characteristics. Under the different imaging conditions, the targets of the same category present some texture features that are not completely consistent. However, the parts of the metal structure contained in the target are always the brighter areas and have the same distribution pattern in the SAR images, which is shown in Fig. 1. Therefore, we can identify aircraft based on the consistency of the scattering characteristics of each category.

To effectively leverage the feature and distribution rules of scattering key points, we extract the stable distribution characteristics of different categories by updating the scattering template online, which can highlight the differences between classes. First, we can get the transformed coordinates $\tilde{C} = \{(\tilde{x}_1, \tilde{y}_1), (\tilde{x}_2, \tilde{y}_2), \ldots, (\tilde{x}_N, \tilde{y}_N)\}$ and the corresponding scattering key features $F_s \in \mathbb{R}^{N \times L \times C_1}$ for each image. Then we design a novel method named scatter grouping algorithm to divide the scattering key points into cluster centers according to their coordinate positions. The cluster centers of each group represent the scattering grouping components of aircraft.

Different from the general algorithm that performs clustering after all the data is acquired, our scatter grouping algorithm updates the new cluster centers in real-time based on the ground-truth labels in each batch. By constantly correcting the cluster centers of different scattering key points in various categories, more accurate results of component positions are obtained. In addition, after getting the predicted component label of each scattering key point, the scattering key feature

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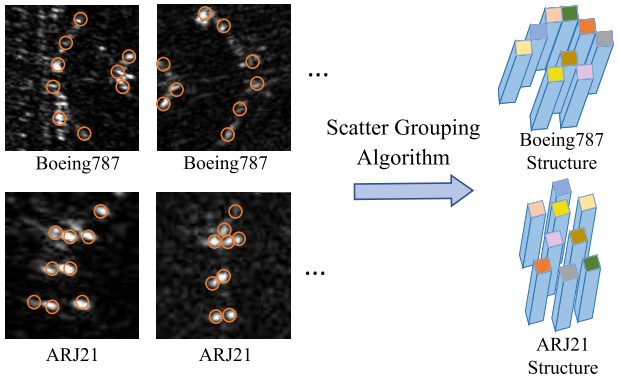


Fig. 5. Process of the scatter grouping algorithm. We divide the scattering key points into cluster centers according to their coordinates and update the corresponding features.

of the corresponding category is updated, which is illustrated in Fig. 5.

The goal of the scatter grouping algorithm is to find the groups $\mathcal{C}_m$ for each category m that minimizes the following objective function over the whole scattering key points set:

$$\mathcal{L} = \sum_{m=1}^{M} \sum_{i=1}^{N} \sum_{\rho \in \mathcal{C}_{mi}} \|\rho - \psi_{mi}\|_2^2 \tag{3}$$

where M is defined as the number of categories, ψ_{mi} denotes ith scatter grouping component information of mth category and $\rho = (\tilde{x}, \tilde{y})$ represents the transformed coordinate in $\tilde{C}$. However, solving $\mathcal{L}$ is an NP-hard problem because finding the optimal solution requires traversing all possible group partitions. Therefore, we design an online grouping method that can be plugged into the overall training process. The final outputs of this method are the refined component information ψ_{mi} and the grouping results $\mathcal{C}_{mi} (1 \le m \le M, 1 \le i \le N)$. Specifically, for any given batch data of F, we can extract the coordinates $\tilde{C}$ of the scattering key points ρ and obtain corresponding features $\hat{F}_s \in \mathbb{R}^{B_m \times N \times C_1'}$ after that F_s is transformed by a linear function, where B_m is the number of targets in the mth category.

For each iteration, we calculate the distance between each point $\rho_{mj} (1 \le j \le N)$ and ψ_{mi} as follows:

$$d_{ji} = \|\rho_{mj} - \psi_{mi}\|_2 \quad \forall i \tag{4}$$

where $\|\cdot\|_2$ is ℓ_2-norm. Then the grouping label λ_j is determined by

$$\lambda_j = \arg\min_{i \in \{1,2,\ldots,N\}} d_{ji} \tag{5}$$

and we can add ρ_{mj} into the corresponding group $\mathcal{C}_{m\lambda_j}$ based on the cluster label λ_j. After grouping the scatter points of the tth batch data, each component information ψ_{mi} is updated based on $\mathcal{C}_{mi}$

$$\psi_{mi}^t = (1 - \alpha_t)\psi_{mi}^{t-1} + \alpha_t \frac{1}{|\mathcal{C}_{mi}|} \sum_{\rho \in \mathcal{C}_{mi}} \rho \tag{6}$$

where α_t is a weight factor. Moreover, $Z_s \in \mathbb{R}^{M \times N \times C_1'}$ is defined as the scattering center features in templates, which is updated by the final grouping label λ_j

$$Z_s(m,i) \leftarrow Z_s(m,i) - \beta_t \Delta Z_s(m,i) \tag{7}$$

where $Z_s(m,i)$ is the slice of Z_s, β_t is the learning rate of Z_s. $\Delta Z_s(m,i)$ can be computed as

$$\Delta Z_s(m,i) = \frac{1}{B_m} \sum_{b_m=1}^{B_m} \frac{\sum_j^N \mathbb{I}(\lambda_j = i)\left(\hat{F}_s(b_m,i) - Z_s(m,i)\right)}{1 + \sum_j^N \mathbb{I}(\lambda_j = i)} \tag{8}$$

where $\hat{F}_s(b_m,i)$ is the slice of $\hat{F}_s$ and $\mathbb{I}(\cdot)$ is the indicator function. The overall process is shown in the Algorithm 1.

Algorithm 1 Scatter Grouping Algorithm

Input: Grouping Max Iteration T, Training Dataset $\mathcal{D}$, Scattering Key Points ρ_{mj}, Scattering Feature $\hat{F}_s$.

for $m = 1, \ldots, M$ **do** {*parallel do with M classes*}
  Initialize group $\mathcal{C}_{mi} = \varnothing, \forall i \in \{1, \ldots, N\}$
  for $t = 1, \ldots, T$ **do**
    Calculate d_{ji} for each point ρ_{mj} based on (4);
    Group ρ_{mj} based on the label λ_j obtained from (5);
    Update ψ_{mi}^t based on (6);
    if $\|\lambda_j^t - \lambda_j^{t-1}\| < \varepsilon, \forall j$ **then**
      Return λ_j^t;
    end if
  end for
  Update Z_s based on λ_j^T and (7);
end for

Output: Grouping Label λ_j^T, Scattering Center Feature Z_s.

Through the online grouping process, we can obtain the coordinates of cluster centers and corresponding scattering center features of various categories. During the inference phase, we utilize the cosine similarity [48] to calculate the similarity scores between the scattering features $\hat{F}_s$ of the test images and scattering center features Z_s of various categories. Then we add the similarity scores to the original model-predicted scores from the feature F_4 for the final classification.

The SAFM constructs a scattering-aware model to weaken the influence of target attitude angles on SAR aircraft classification. Besides, it utilizes the online scatter grouping algorithm to obtain the statistical positions and the corresponding scattering semantic features of different components.

C. Scattering Topology Module (STM)

Due to the discreteness of aircraft scattering points, it is easy to ignore the correlation between various components, which leads to insufficient spatial features and the wrong classification of targets. Inspired by the topology and the skeleton characteristics of the aircraft, we utilize the topology nodes and topology edges to represent the scattering key points and their spatial relations, respectively. Then we propose a novel scattering topology model to construct the geometric relationship from the spatial and semantic features, which is shown in Fig. 6. Furthermore, considering the consistency of the scattering topologies, we calculate the Kullback-Leibler (KL)

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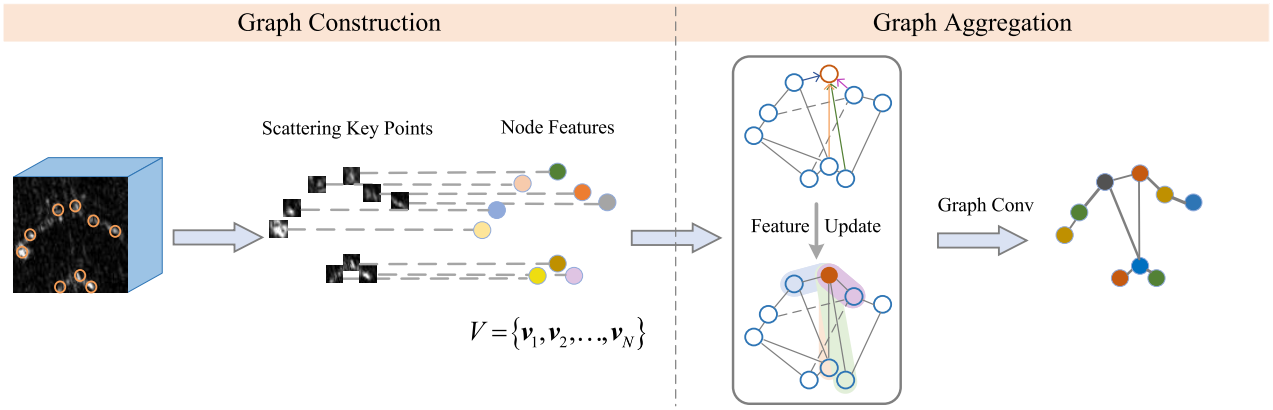


Fig. 6. Framework of the STM. It contains two parts: 1) graph construction and 2) graph aggregation. In the graph construction, we use the topology nodes and topology edges to represent the scattering key points and their spatial relations, respectively. The final scattering topology structure is obtained by updating the graph network and aggregating features.

divergence to reduce the difference in the intraclass topology. While strengthening the connection of scattering key points, the scattering characteristic structures of different categories are highlighted.

1) Graph Construction: The positional and semantic information of different scattering points is vital to explore the spatial structure of the targets in SAR images. We define the discrete scattering key points as the topology nodes, and $V = \{\boldsymbol{v}_1, \boldsymbol{v}_2, \ldots, \boldsymbol{v}_N\}$ represents a set of node features. The edge of the node feature $\boldsymbol{v}_i$ and $\boldsymbol{v}_j$ is denoted as $e_{ij} \in E, \forall i, j \in \{1, \ldots, N\}$, where $e_{ij} = 1$. Then, the graph $G = \{V, E\}$ is utilized to model the spatial relations of the input scattering key features F_s.

The edge $e_{ij} \in E$ corresponds to the relations between any two point elements. Different from the general adjacency matrix, we supplement the location information $\tilde{C} = \{(\tilde{x}_1, \tilde{y}_1), (\tilde{x}_2, \tilde{y}_2), \ldots, (\tilde{x}_N, \tilde{y}_N)\}$ of the scattering key points to generate the weighted adjacency matrix $A \in \mathbb{R}^{N \times N}$. Specifically, we use a polar function $\gamma(i, j)$ to reflect the pairwise spatial connection between the node i and node j. This process is calculated as

$$\begin{aligned} \gamma(i, j) &= (\ell_{ij}, \phi_{ij}) \\ \ell_{ij} &= \sqrt{(\tilde{y}_j - \tilde{y}_i)^2 + (\tilde{x}_j - \tilde{x}_i)^2} \\ \phi_{ij} &= \arctan\left(\frac{\tilde{y}_j - \tilde{y}_i}{\tilde{x}_j - \tilde{x}_i}\right). \end{aligned} \tag{9}$$

After getting the relation function $\gamma(i, j)$ between different nodes, a weighted operator is needed to describe the influence between nodes in the graph structure. Inspired by the mixture model network (MoNet) [49], we utilize K Gaussian kernels as the weighted operator with the learnable mean vectors μ_k and covariance matrixs Σ_k, which are computed as

$$w_k(\gamma(i, j)) = \exp\left(-\frac{1}{2}(\gamma(i, j) - \mu_k)^T \Sigma_k^{-1} (\gamma(i, j) - \mu_k)\right)$$

where the Gaussian kernel $w_k(\cdot)$ represents the spatial information of the scattering key points and is the entry value of the weighted adjacency matrix A. This Gaussian process facilitates the network to adaptively learn the spatial and semantic relationship between different scattering key points.

2) Graph Aggregation: After completing the graph network construction $G = \{V, E\}$ in the previous step, the graph convolution is adopted to propagate and update neighbor node features. Formally, given the scattering feature input $\boldsymbol{v}_j \in V$ and the Gaussian weight $w_k(\gamma(i, j))$, the representation for the node i is aggregated by its neighborhood

$$\boldsymbol{z}'_k(i) = \sum_{j \in \text{Neighbor}(i)} w_k(\gamma(i, j)) \boldsymbol{v}_j e_{ij} \tag{10}$$

where $\boldsymbol{z}'_k(i)$ is a weighted fusion of the neighbor features and Neighbor(i) contains all nodes in our setting. After getting K features, we perform the feature fusion to get $\boldsymbol{z}'_i$ by $\boldsymbol{z}'_i = \sum_{k=1}^{K} \boldsymbol{z}'_k(i)$. Then the feature $\boldsymbol{z}'_i$ goes through a linear transformation $\boldsymbol{L}$ to obtain the converted feature $\boldsymbol{h}_i = \boldsymbol{L}(\boldsymbol{z}'_i)$.

After obtaining feature $\boldsymbol{z}'_i$ of each scattering key point, the structural feature $\boldsymbol{z}' = [\boldsymbol{z}'_1, \boldsymbol{z}'_2, \ldots, \boldsymbol{z}'_N]$ is performed by the element-wise addition with F'_2 according to the coordinates to generate the feature F''_2. Then, this feature is concatenated to F'_2 on the channel dimension for further classification.

In order to improve the compactness of the features within the class, we exploit the KL divergence to minimize the gap of scattering key features within the same category. The KL divergence is adopted to measure the similarity as

$$D_{\text{KL}} = \text{KL}(P||Q) = \int p(x) \log \frac{p(x)}{q(x)} dx \tag{11}$$

where P and Q are two different distribution representations. $p(x)$ and $q(x)$ are the corresponding probability density function over variable x. Specifically, after obtaining the feature $\boldsymbol{h}_i$ of each sample, we minimize the KL divergence of different sample features $\boldsymbol{h}_i$ in the same category to optimize the network parameter, which is conducive to maintaining the scattering characteristics from same category consistent.

The STM takes the coordinates and distances between the scattering key points into account and constructs the spatial topology relationship between them. The aggregated features combine the scattering topology structure of the targets and represent the spatial relationship between scattering key points.

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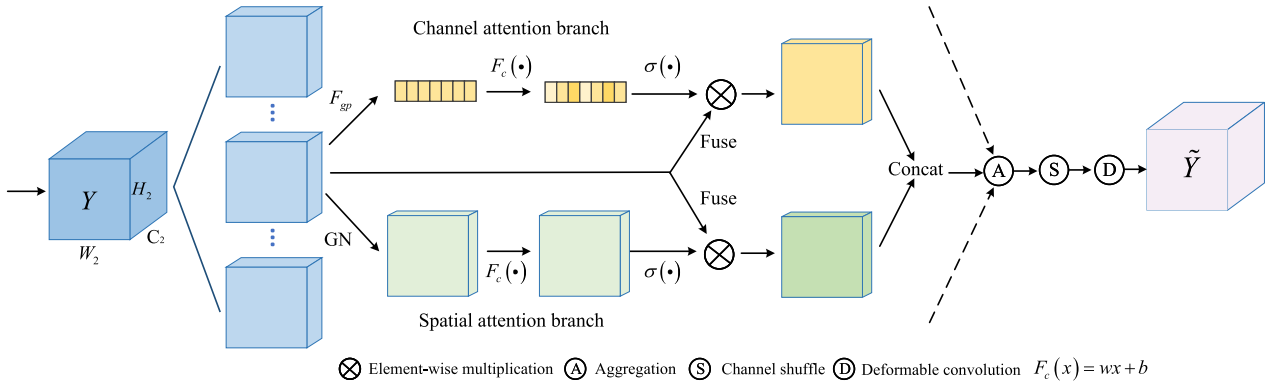


Fig. 7. Structure of the proposed CAE module. The CAE first split the overall feature into sub-features along the channel dimension parallelly. Each sub-feature goes through both the channel and spatial attention branches simultaneously. Then two branches are concatenated to obtain the sub-features, and all sub-features are aggregated to generate the whole features. Finally, the fused feature is adopted by the channel shuffle operator and deformable convolution to enhance information interaction.

This operation is beneficial to classify SAR images more effectively.

D. Context Attention Excitation (CAE)

Due to the limitations of the size and fixed shape of the convolution kernel, the general convolution operation ignores the global information, which easily leads to the misclassification of targets with a similar local feature. In addition, the influence of speckle noise causes the lack of prominence of aircraft targets. Inspired by the convolutional block attention module (CBAM) [50] and global context network (GCNet) [51], we propose the CAE to enhance the context information adaptively and improve the discrimination ability.

The proposed CAE framework is presented in Fig. 7. First, we divide the overall feature map into several sub-features along the channel dimension, which can increase the calculation speed. Then the sub-features are enhanced with the channel and spatial attention parallelly. After that, we aggregate all sub-features for rich global information and adopt the channel shuffle [52] to strengthen the communication between different channels.

Specifically, we define the given feature map as $F_2 \in \mathbb{R}^{W_2 \times H_2 \times C_2}$, where W_2, H_2, and C_2 represent the spatial width, height and channel number of F_2, respectively. First, the CAE groups F_2 into G sub-features in terms of channel dimension, i.e., $Y = [Y_1, \ldots, Y_G]$, $Y_k \in \mathbb{R}^{W_2 \times H_2 \times C_2/G}$ is the kth element. The sub-feature gradually represents a specific characteristic information as the training progresses. Then, the feature Y_k is passed through the channel and spatial attention branches to capture the context information simultaneously.

In the channel attention branch, we enhance the full interaction of different channels of feature maps. First, the global average pooling is adopted to compress the spatial dimension to get global information $a_1 \in \mathbb{R}^{1 \times 1 \times C_2/G}$, which is obtained by

$$a_1 = F_{gp}(Y_k) = \frac{1}{W_2 \times H_2} \sum_{i=1}^{W_2} \sum_{j=1}^{H_2} Y_k(i, j). \tag{12}$$

Then the adaptively adjusted weight of different channels is designed to reflect the correlation between them. Then the different weight coefficients are multiplied by the corresponding original features to produce the calibrated information, which is computed by the gating mechanism with the sigmoid activation

$$Y_{k1} = \sigma(F_c(a_1)) \otimes Y_k = \sigma(w_1 a_1 + b_1) \otimes Y_k \tag{13}$$

where $w_1 \in \mathbb{R}^{1 \times 1 \times C_2/G}$ and $b_1 \in \mathbb{R}^{1 \times 1 \times C_2/G}$ are the scaling parameters in the channel attention branch, and $\otimes$ represents the element-wise multiplication, and σ is the activation function, i.e., rectified linear unit (ReLu). Compared with the SE-Net [53], we reduce the parameters and achieve a good balance between the classification effect and speed. By explicitly modeling the features among different channels, we can highlight the important features while suppressing the different background clutter noise.

In addition to the effective connection in the channel attention branch, we supplement spatial attention by focusing on the important location. Inspired by the group norm (GN) [54], we utilize it to obtain the preliminary spatial statistics. Then $F_c(\cdot)$ is also applied to capture the long-range dependency. The weighted output is calculated as

$$a_2 = \mathrm{GN}(Y_k) \tag{14}$$

$$Y_{k2} = \sigma(F_c(s_2)) \otimes Y_k = \sigma(w_2 a_2 + b_2) \otimes Y_k \tag{15}$$

where $w_2 \in \mathbb{R}^{1 \times 1 \times C_2/G}$ and $b_2 \in \mathbb{R}^{1 \times 1 \times C_2/G}$ are the scaling parameters in the spatial attention branch. After obtaining the two feature maps in the two branches, we perform a concatenation operation to fuse Y_{k1} and Y_{k2} to get the new sub-feature Y'_k. Moreover, all new sub-features are aggregated, and the channel shuffle is used to enhance the interaction between various sub-features, which generates the output Y'.

Considering the limitation of the fixed shape of the convolution kernel, we utilize the deformable convolution [55] on the Y' with the offsets to classify the aircraft exactly, which is different from the article [56]. This process is denoted as

$$\tilde{Y}(p_0) = \sum_{p_n \in R} w(p_n) \cdot Y'(p_0 + p_n + \Delta p_n) \tag{16}$$

where $\Delta p_n (n = 1, \ldots, |R|)$ is the kernel offset. For the input feature map Y', a 3×3 kernel R with dilation 1 is

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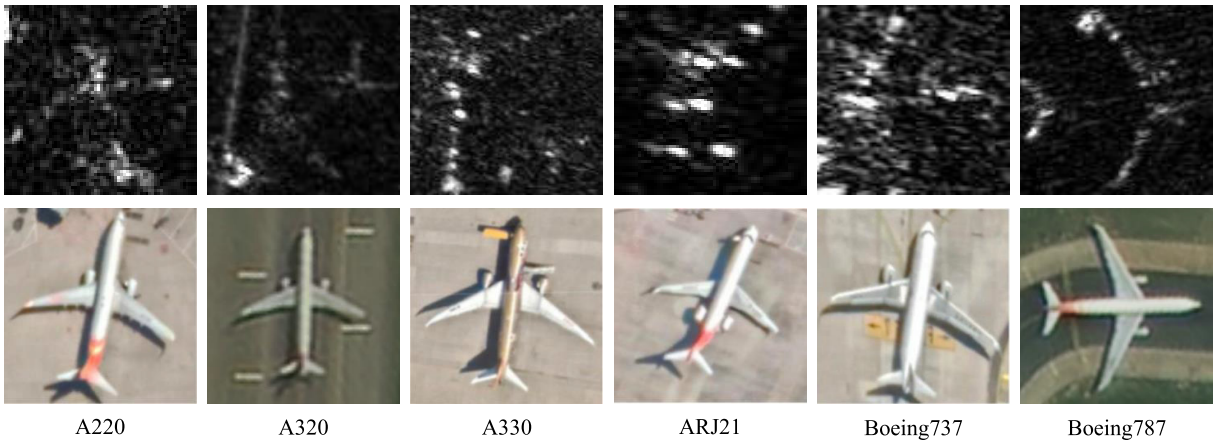


Fig. 8. Examples of different categories of civil aircraft in the dataset SAR-ACD.

used to sample over it. The refined output of each location p_0 is computed by the weighted sampling values in R. Finally, we can get the refined feature $\tilde{Y}$ for further classification.

In terms of target classification, the CAE module combines with global perceptions, which is beneficial to increase the distinction from background interference. Besides, it can adjust the sizes of the receptive fields and capture the long-range dependency of features. The contextual information is enhanced to improve classification accuracy.

E. Objective Function

The total training loss of ST-Net contains three parts: 1) the matching loss of SAFM; 2) the topology loss of the STM; and 3) the classification loss of the CAE.

The SAFM module is optimized by the matching loss [57], which is calculated by the grouping labels. For each image, we first obtain the label $\Lambda = [\lambda_1, \ldots, \lambda_N]$ of its scattering points. Then the matching loss can be written as

$$L_{\text{mat}} = \frac{1}{BN} \sum_{b=1}^{B} \sum_{i=1}^{N} \left\| \hat{F}_s(b, \Lambda_i) - Z_s(y_b, \Lambda_i) \right\|_2^2 \quad (17)$$

where y_b denotes the ground-truth category label of bth sample.

To embody the structural differences of the same category, we leverage the KL to compute the topology loss L_{topo} based on (11).

In terms of the classification loss, the cross-entropy [58] is applied for evaluating the distance between the predicted labels and the true labels, which is calculated as

$$L_{\text{cls}} = -\frac{1}{B} \sum_{b=1}^{B} \sum_{m=1}^{M} y_{bm} \log(p_{bm}) \quad (18)$$

where p_{bm} represents the probability values. The value of y_{bm} is defined as 1 if the ith aircraft is correctly classified to class m, otherwise it is 0. Finally, the loss function of ST-Net can be formulated as follows:

$$L = L_{\text{cls}} + \lambda_{\text{mat}} L_{\text{mat}} + \lambda_{\text{topo}} L_{\text{topo}} \quad (19)$$

where λ_{mat} and λ_{topo} are the regularization factors of L_{mat} and L_{topo}, respectively.

IV. Experiments

A. Dataset

We conduct extensive experiments on the SAR-ACD [15] to assess the effectiveness of the proposed network. The SAR-ACD contains 11 C-band images from the Gaofen-3 satellite working in the spotlight mode. These images of different time phases cover the Shanghai Hongqiao Airport area and other airports. The SAR-ACD consists of 4322 aircraft clips with six civil aircraft categories and 14 other aircraft categories. Based on the corresponding optical images, this dataset is manually labeled by combining expert interpretation. Fig. 8 presents some example SAR clips of civil aircraft and the corresponding optical images, containing the A220, A320, A330, ARJ21, Boeing737, and Boeing787. Meanwhile, the original number of civil aircraft in each category is illustrated in Fig. 9. In terms of the data expansion, we utilize the sharpness, brightness, and contrast transformation to alleviate the problem of an imbalanced number of targets. The rotation and reversal are not adopted because the imaging results of SAR targets change under different target attitude angles.

The constructed SAR-ACD includes rich fine-grained types and diverse sizes, which conform to the real situation in practical applications. In addition, this dataset contains multiple airport scenes under different imaging conditions, and there is a certain amount of clutter interference. Thus, SAR-ACD is a challenging dataset aimed at aircraft classification. Part of this dataset is available online.[1]

B. Implementation Detail

We utilize the ResNet-34 [44] as the backbone in the overall structure, which contains the residual structure and can alleviate the vanishing gradients problem caused by network deepening. There are a total of 32 images per minibatch to train this model. The stochastic gradient descent (SGD) [59] is leveraged as the optimizer with a momentum of 0.9. In the training process, the learning rate is set to 0.001 initially and decreases to 0.0001 at the 40th epoch for the convergence of

[1] [Online]. Available: https://github.com/AICyberTeam/SAR-ACD

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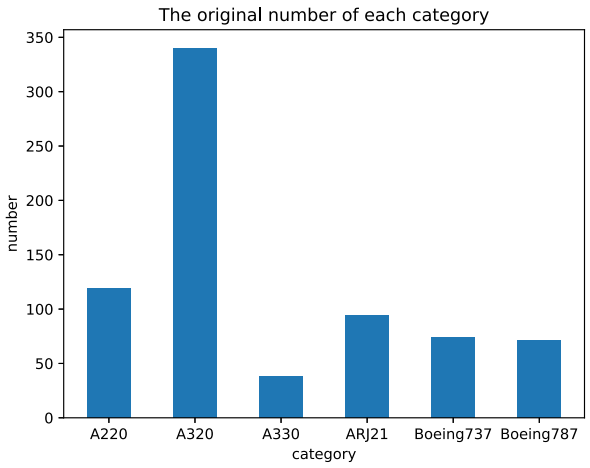


Fig. 9. Original distribution number of each category of civil aircraft.

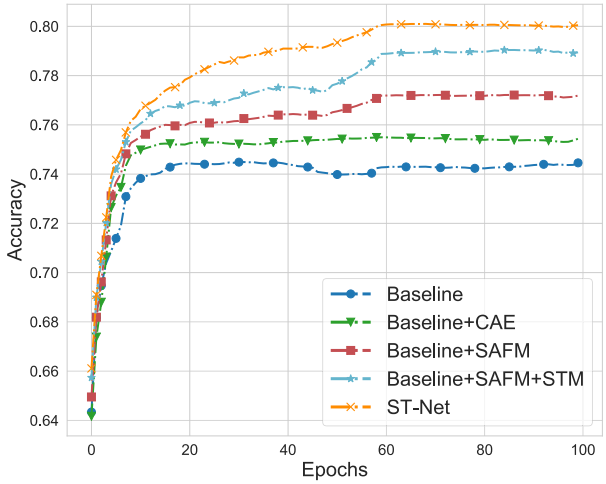


Fig. 10. Accuracy curves of different components in the ST-Net.

loss. The training and testing experiments are implemented on a 16-GB NVIDIA Tesla P100 GPU environment for 100 epochs.

C. Evaluation Metrics

To quantify the performance of the aircraft classification task, we utilize the overall accuracy (OA), confusion matrix, and kappa coefficient to access the image prediction accuracy. The Kappa focuses on consistency to alleviate the problem of unbalanced quantity of different categories. It is computed by OA and confusion matrix as follows:

$$\begin{aligned} \mathrm{OA} &= \frac{\sum_{m=1}^{N_c} N_m}{N_{\mathrm{all}}} \\ p_e &= \frac{1}{N_{\mathrm{all}}^2} \sum_{m=1}^{N_c} \left(\mathrm{gt}_m \times \mathrm{pred}_m\right) \\ \mathrm{Kappa} &= \frac{\mathrm{OA} - p_e}{1 - p_e} \end{aligned} \tag{20}$$

where N_m, N_{all}, N_c represent the number of correctly classified numbers of mth category, the number of test images and the number of all categories, respectively. gt_m and pred_m are defined as the ground truth number and the predicted number of mth category, respectively. In addition, to calculate the confusion matrix, we used the best model in the training process.

D. Ablation Experiments

In this part, extensive experiments are implemented based on the SAR-ACD to evaluate the effectiveness of each component in ST-Net. For a fair comparison, all subsequent studies are performed with the same settings. The overall comparisons are displayed in Table I and the accuracy curves are presented in Fig. 10. It can be seen that different components bring improvement to the network to a certain extent. Compared with the baseline, the final model achieves 3.9% and 6.0% higher in terms of 50% and 80% training datasets, respectively. In addition to the upgrading of accuracy, there is also a great improvement in the Kappa coefficient, which reflects the satisfactory results of the proposed method. Furthermore, the classification results of some instances presented in Table II also illustrate the superiority of different modules.

TABLE I
INFLUENCE OF EACH MODULE IN ST-NET. BEST RESULTS ARE BOLDED

Modules			OA(%)		Kappa	
SAFM	STM	CAE	50%	80%	50%	80%
×	×	×	71.40	74.30	0.6970	0.7346
✓	×	×	73.20	76.82	0.7210	0.7533
×	✓	×	73.86	77.31	0.7275	0.7645
×	×	✓	72.54	75.43	0.7212	0.7421
✓	✓	×	74.65	79.49	0.7362	0.7814
✓	✓	✓	**75.31**	**80.34**	**0.7438**	**0.7923**

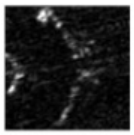
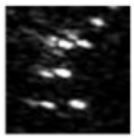
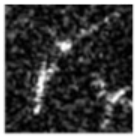
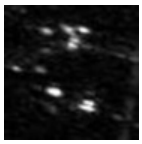
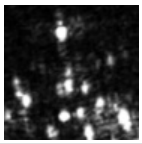
Next, we will analyze the effectiveness of the SAFM containing SAR scattering characteristics. Furthermore, the STM with the spatial relationship and CAE with fused features are studied in detail.

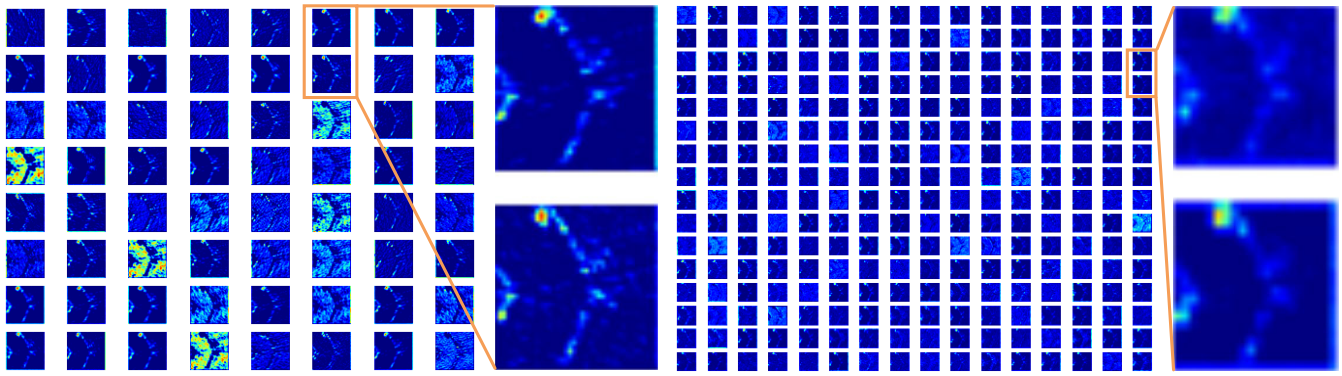
1) Effect of SAFM: We first analyze the ResNet structure with different layers in the backbone. The ResNet structure adopts a skip connection method, which can effectively avoid gradient explosion. We conduct the experiments of ResNet-18, ResNet-34, and ResNet-50 on the SAR-ACD. The corresponding results in Table III show that the accuracy and Kappa are optimal when the number of layers is specified as 34. The features learned by ResNet-18 are insufficient, and ResNet-50 is prone to over-fitting due to its deep network. Therefore, we choose ResNet-34 as the baseline for better effect.

As described in Section III-B, the scattering features are extracted from the feature map to enhance the distribution relationships of different scattering points. We explore the various locations of adding the SAFM, and the corresponding results are shown in Table IV. We can observe that adding SAFM to the position after feature map F_1 can obtain the highest accuracy, so we choose to arrange the SAFM after F_1. In addition, we make a comparison of feature maps F_1 and F_2. The visualization of all the channels of F_1 and F_2 is illustrated in Fig. 11. It shows that low-level feature F_1 contains more

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TABLE II
CLASSIFICATION RESULTS OF SOME AIRCRAFT PATCHES OF ST-NET

Images	True Labels	Baseline	With only SAFM	With only STM	With only CAE	ST-Net
	Boeing787	A220 ×	Boeing787 ✓	Boeing787 ✓	Boeing787 ✓	Boeing787 ✓
	ARJ21	A220 ×	ARJ21 ✓	ARJ21 ✓	A220 ×	ARJ21 ✓
	A220	Boeing787 ×	A220 ✓	Boeing737 ×	A220 ✓	A220 ✓
	Boeing737	ARJ21 ×	Boeing737 ✓	A220 ×	A220 ×	Boeing737 ✓
	A320	Boeing737 ×	A220 ×	Boeing737 ×	Boeing737 ×	A320 ✓



(a) (b)

Fig. 11. Comparison visualization of all the channels of feature map F_1 and F_2. Both the first and second rows are the same aircraft Boeing787. (a) Representation characterization of 64 channels in F_1. (b) Representation characterization of 256 channels in F_2.

detailed information, and the texture features and structural contours of the aircraft are more prominent. The scattering features extracted from F_1 maintain a good balance between detail and semantic information. It is beneficial to strengthen the connection between scattering points and enhance the distinguishable features.

To clearly present the effect of the SAFM, the qualitative results using t-distributed stochastic neighbor embedding (t-SNE) [60] are shown in Fig. 12. In the t-SNE visualization, the farther the distance between different categories, the better the recognition accuracy of the model. Fig. 12(a) and (b) illustrate the visualization without and with the addition of SAFM based on the baseline, respectively. We can observe that there are many confusions in the feature embedding of some categories in Fig. 12(a). After supplementing the SAFM, the classification error is alleviated and the feature embeddings from the same class are more aggregated, which is shown in Fig. 12(b). SAFM separates the features

TABLE III
PERFORMANCE OF THE DEEP RESIDUAL NETWORKS. BEST RESULTS ARE BOLDED

Methods	OA(%)		Kappa		Params(M)
	50%	80%	50%	80%	
ResNet-18	70.31	73.42	0.6784	0.7213	**11.55**
ResNet-34	**71.40**	**74.30**	**0.6970**	**0.7346**	21.28
ResNet-50	69.38	73.82	0.6732	0.7232	23.91

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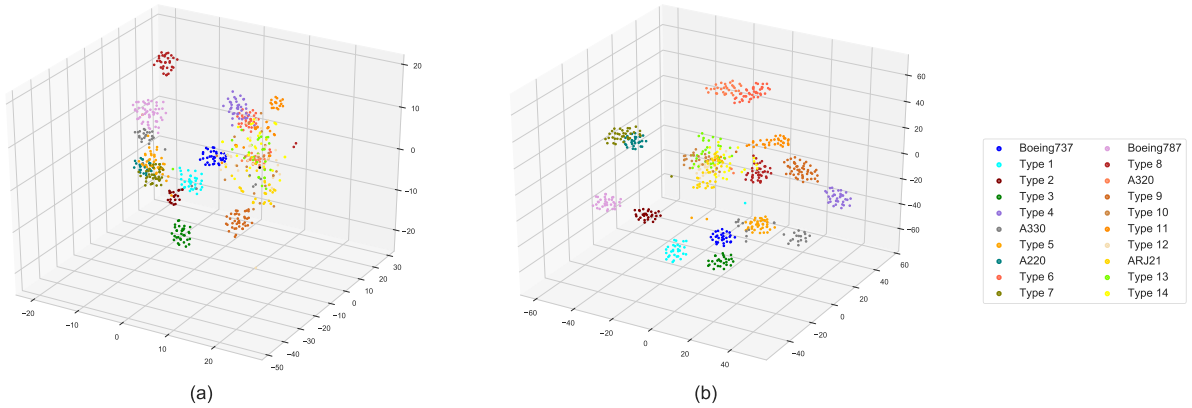


Fig. 12. Visualization of the feature space using t-SNE for all categories with different colors. (a) Baseline. (b) Baseline with SAFM.

TABLE IV

INFLUENCE OF CHANGING THE LOCATIONS OF ADDING THE SAFM. BEST RESULTS ARE BOLDED

Feature Map	OA(%)		Kappa	
	50%	80%	50%	80%
F_1	**73.20**	**76.82**	**0.7210**	**0.7533**
F_2	72.85	76.26	0.7179	0.7495
F_3	72.31	75.79	0.7114	0.7423

TABLE V

EFFECTS OF DIFFERENT REGULARIZATION FACTORS. BEST RESULTS ARE BOLDED

λ_{mat}	0.01	0.1	1	10
OA(%)	74.56	**80.34**	0.7731	0.7233
Kappa	0.7286	**0.7923**	0.7579	0.7045
λ_{topo}	0.05	0.2	1	10
OA(%)	75.20	**80.34**	78.47	75.33
Kappa	0.7389	**0.7923**	0.7719	0.7423

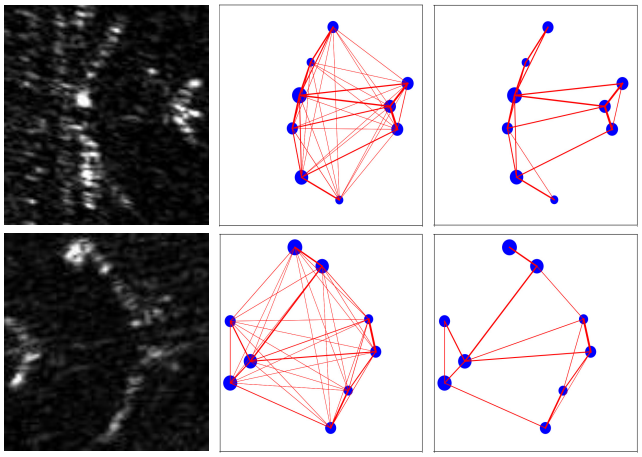


Fig. 13. Topology visualization of aircraft. The solid circles and lines represent nodes and edges, respectively. The size of the radius and the thickness of the line reflect the value of the feature and the weight of the edge, respectively. The aircraft in the first and second rows are Boeing787. (a) Original images. Visualization of (b) all edges with STM and (c) edges whose weights exceed a threshold.

between different categories further, and the intraclass features of the same category are closer together. These results indicate that representation with scattering characteristics has a critical impact on the improvement of classification accuracy.

2) Effect of STM: To enhance the spatial relationships between the scattering key points, we leverage the GCN to construct the scattering-aware topology module. Compared with the baseline, the STM increases the classification accuracy of 50% and 80% training ratios by 2.4% and 3.0%, respectively. In addition, the Kappa coefficient has also been improved to a certain extent, which demonstrates the excellent performance of the STM. To visualize the topology of aircraft, we compute the summation of each scattering key feature along the channel dimension. As shown in Fig. 13, the solid circles and lines represent nodes and edges, respectively. The size of the radius reflects the values of the features. The larger the radius, the larger the value of the feature, and the more obvious the feature. For edges in the graph structure, the weight of the obtained edge is represented by the lines with different thicknesses. The thicker the line, the closer the relationship between the corresponding nodes. As illustrated in Fig. 13(b), the topology structure of the Boeing787 aircraft embodies the relationships between the scattering points. The structure after removing edges with weights less than the threshold is shown in Fig. 13(c). We can observe that the two targets from the same category have similar topologies, which is conducive to improving classification accuracy. Besides, more spatial information is aggregated by the feature updating of neighbor nodes. Overall, the proposed STM combines the positional relationship between scattering points to construct a topology representation, which contributes to the fine-grained classification task.

3) Effect of CAE: The proposed CAE is introduced to aggregate the global feature for highlighting important features and suppressing the clutter interference in the background. The results in Table I show that the CAE module achieves 1.1% and 1.2% improvement under the 50% and 80% training ratios compared with the baseline, respectively. In addition, CAE

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TABLE VI
RESULTS OF DIFFERENT CNN-BASED ALGORITHMS ON THE SAR-ACD. BEST RESULTS ARE BOLDED

Methods	OA(%)		Kappa		Params (M)	FLOPs (G)
	50%	80%	50%	80%		
AlexNet [61]	69.23	75.90	0.6784	0.7456	23.40	0.05
VGG16 [62]	73.23	76.16	0.7196	0.7497	40.31	1.28
GoogleNet [63]	72.67	74.63	0.7128	0.7314	6.32	2.14
Inception-v3 [64]	72.30	76.32	0.7102	0.7502	22.31	15.60
SE-ResNext-50 [53]	72.21	77.39	0.7089	0.7603	33.06	0.48
ShuffleNet v2 [52]	69.71	74.30	0.6803	0.7295	2.50	0.38
EfficientNet-B0 [65]	70.25	75.33	0.6918	0.7398	**1.96**	**0.004**
MMAL-Net [66]	73.69	77.62	0.7227	0.7613	22.37	5.98
SimAM [67]	74.51	78.31	0.7314	0.7709	25.03	4.31
MKSFF-CNN [68]	72.93	75.36	0.7154	0.7392	11.48	0.69
ST-Net	**75.31**	**80.34**	**0.7438**	**0.7923**	23.91	7.85

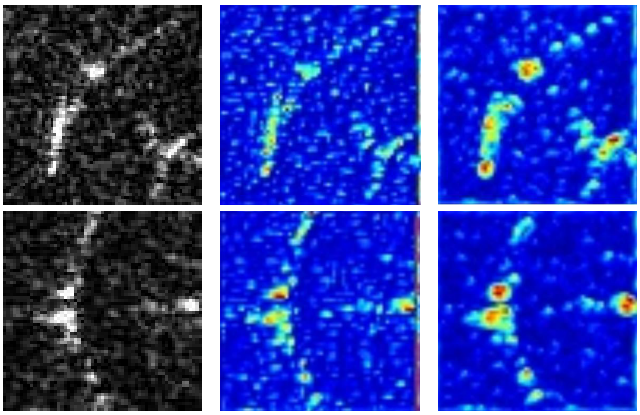


Fig. 14. Visualization of the features of different aircrafts. (a) Original images. Visualization of feature map F_2 (b) in baseline and (c) of the model with CAE.

leads to a notable increase of 2.5% in Kappa with 50% training ratio. The growth of indicators demonstrates the advance of the proposed CAE structure. In addition, we analyze the balancing parameters in the loss. The total loss of our method in (20) is weighted by two regularization factors λ_{mat} and λ_{topo}. The influence of λ_{mat} and λ_{topo} are revealed in Table V. When the λ_{mat} and λ_{topo} are fine-tuned to 0.1 and 0.2, respectively, the classification result reaches the best.

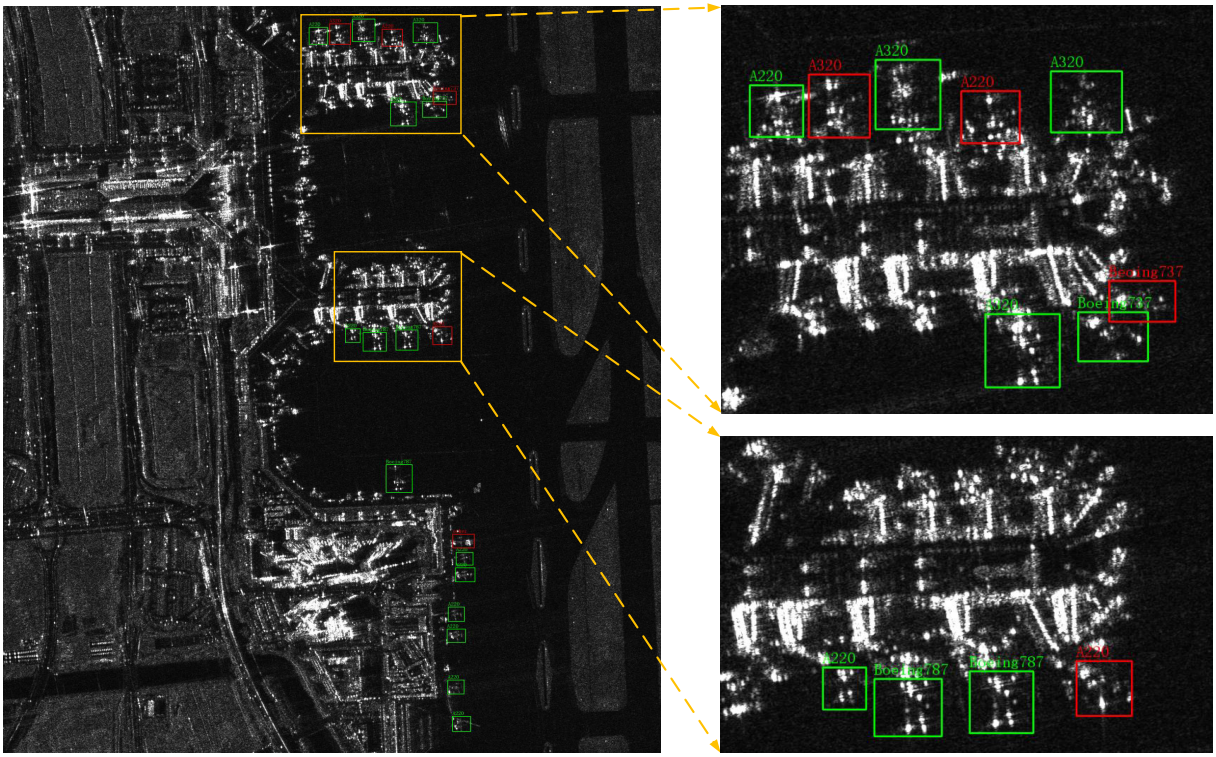
For intuitive comparison, we present some qualitative visualization results of the feature map F_2 without and with CAE in Fig. 14. The activation heatmap of the extracted feature is the summation of the values in each row along the channel dimension. Both original SAR images and visualized heatmaps are scaled to a uniform size for more effective display. As illustrated in Fig. 14(b), there are many noise and an inconspicuous target in the baseline because of the clutter interference. To deal with this problem and capture more information, we propose the CAE to enhance the contextual connections by adding different weights in the channel layer. As shown in Fig. 14(c), the model with CAE pays more attention to the distinguishable regions, so these important parts have high activation scores. At the same time, the influence of the noise is reduced and the background becomes pure. It proves that CAE effectively enhances the saliency of the targets and reduces interference in the complex background. These noticeable improvements with CAE may be attributed to two aspects. First, global features are aggregated in a channel group and adaptive fashion, which provides the context and semantics informative in terms of channels and spatial for classification. Second, combining the long-range dependencies and the deformable convolution, the model takes full advantage of the effective communication between different parts and highlights the discriminative representation.

Through the above analysis and description, we have verified the effectiveness of the proposed three modules. It turns out that the SAFM and STM modules reduce misclassification by combining scattering characteristics and constructing the topology of different categories. Furthermore, the CAE improves classification accuracy by enhancing contextual information.

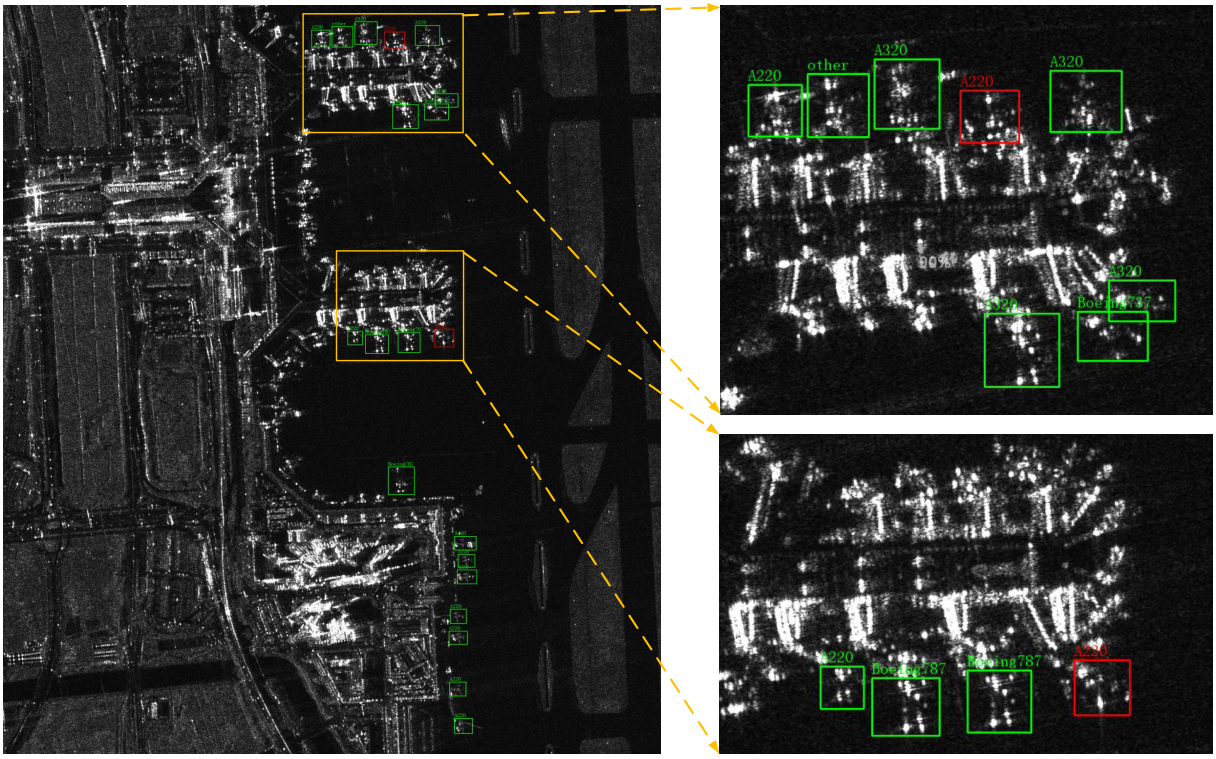
E. Comparison With State-of-the-Art Methods

To evaluate the performance of the proposed method comprehensively, we compare the ST-Net with some CNN-based methods on SAR-ACD. Considering the diversity of methods, we choose ten different methods, i.e., AlexNet [61], visual geometry group with 16 weight layers (VGG16) [62], GoogLeNet [63], Inception-v3 [64], SE-ResNext-50 [53], ShuffleNet v2 [52], EfficientNet-B0 [65], multi-branch and multi-scale learning network (MMAL-Net) [66], SimAM [67] and multikernel-size feature fusion (MKSFF)-CNN [68] for comparative experiments, which cover some current application of deep learning to target recognition. As an algorithm for classifying the SAR targets, MKSFF-CNN abandons the fixed-size kernels and uses the dilated convolution to extract the features, which fuses the information from various sizes of convolutional kernels. The experimental results of different methods are displayed in Table VI. It is evident that our method achieves the OA of 75.31% and 80.34% on SAR-ACD under 50% and 80% ratios, respectively. Benefitting from the SAR topology representation and scattering feature, the ST-Net outperforms other competitive methods and achieves the best performance in classification accuracy, which demonstrates its superiority. In addition to the accuracy, we also compute the Params and FLOPs of different methods for the comprehensive comparison. The EfficientNet-B0 carries

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(a)



(b)

Fig. 15. Comparison classification results of the SimAM and ST-Net on the large scene. The green and red represent the correct classification and misclassification, respectively. (a) SimAM. (b) Proposed ST-Net.

minimal memory consumption and the highest computational efficiency. Besides, VGG16 and SE-ResNext-50 increase the number of parameters while improving the performance. Compared with these methods, the proposed ST-Net maintains a certain balance between computational efficiency and classification accuracy.

To reflect the effectiveness and practical applicability of ST-Net, we verify the proposed algorithm on large-scale

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images. In the above ten comparison methods, SimAM has the best performance except for ST-Net by optimizing the energy function. Thus, we visualize the classification results of SimAM and ST-Net on a SAR image of Hongqiao Airport, Shanghai City, China. The detailed comparison results in Fig. 15 show that the vast majority of targets in this area are correctly classified and the category confusion is reduced in the proposed method, which confirms the appreciable improvement of ST-Net. Compared with SimAM and other methods, the proposed framework has the following advantages. First, benefitting from the scattering key features extracted from aircraft in SAR images, ST-Net can alleviate the influence of target attitude angles and improve classification accuracy. Second, the scattering-aware topology structure enhances the geometric representation and spatial relations between the discrete aircraft components, which reduces the misclassification effectively. Third, equipped with the ability to enhance long-range information and enrich the global feature, the proposed method can highlight the important but unobvious components of targets. In summary, the ST-Net achieves the SOTA performance and exhibits excellent robustness in classification tasks.

Combined with the representative features and context information, the classification performance of ST-Net has been greatly improved. However, there are still some misclassified targets, as illustrated by the red rectangles in Fig. 15(b). Due to the general rules of aircraft docking, some components of targets are often integrated with ground facilities and terminal buildings. This phenomenon leads the network to regard the strong scattering interference in the background as a part of the target, which affects the class determination and causes category confusion. Motivated by the article in the task of detection [69], [70], we can consider a related method with the spatial information and investigate a more effective feature for the scattering topology relations to further highlight the differences between targets and interference in the future work.

V. Conclusion

In this article, a novel framework named ST-Net is designed for aircraft classification in high-resolution SAR images. Compared with the existing classification methods, ST-Net takes advantage of scattering characteristics and constructs the graph-based topology relationships. First, SAFM extracts and leverages the scattering feature of various categories to guide the network to learn the discriminative features, which is beneficial to mitigate the effect of imaging variability in SAR images. Second, STM is introduced to enhance the spatial topology relations of the various components and aggregate the geometric representation adaptively. Third, CAE can capture significant context information and strengthen the discriminative features while suppressing background interference. We construct a public dataset SAR-ACD, and the satisfactory results of ablation studies on it verify the effectiveness of the proposed modules. In contrast with the commonly used CNN-based methods, the classification results and outstanding performance demonstrate the superiority and stability of the proposed ST-Net.

References

[1] Q. Guo, H. Wang, and F. Xu, "Scattering enhanced attention pyramid network for aircraft detection in SAR images," *IEEE Trans. Geosci. Remote Sens.*, vol. 59, no. 9, pp. 7570–7587, Sep. 2021.

[2] M. Yang, X. Bai, L. Wang, and F. Zhou, "Mixed loss graph attention network for few-shot SAR target classification," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 60, pp. 1–13, 2022.

[3] J. Chen et al., "Blind NCS-based autofocus for airborne wide-beam SAR imaging," *IEEE Trans. Comput. Imag.*, vol. 8, pp. 626–638, 2022, doi: 10.1109/TCI.2022.3194745.

[4] J. Chen et al., "Blind NCS-based autofocus for airborne wide-beam SAR imaging," *IEEE Trans. Comput. Imag.*, vol. 8, pp. 626–638, 2022, doi: 10.1109/TCI.2022.3194745.

[5] S. Zhao, Z. Zhang, W. Guo, and Y. Luo, "An automatic ship detection method adapting to different satellites SAR images with feature alignment and compensation loss," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5225217.

[6] X. Sun et al., "FAIR1M: A benchmark dataset for fine-grained object recognition in high-resolution remote sensing imagery," *ISPRS J. Photogramm. Remote Sens.*, vol. 184, pp. 116–130, Feb. 2022.

[7] J. Kang, Y. Wang, and X. X. Zhu, "Multipass SAR interferometry based on total variation regularized robust low rank tensor decomposition," *IEEE Trans. Geosci. Remote Sens.*, vol. 58, no. 8, pp. 5354–5366, Aug. 2020.

[8] H. Jing, Z. Wang, X. Sun, D. Xiao, and K. Fu, "PSRN: Polarimetric space reconstruction network for PolSAR image semantic segmentation," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 14, pp. 10716–10732, 2021.

[9] J. Chen, B. Zhang, and C. Wang, "Backscattering feature analysis and recognition of civilian aircraft in TerraSAR-X images," *IEEE Geosci. Remote Sens. Lett.*, vol. 12, no. 4, pp. 796–800, Apr. 2015.

[10] H. Ruohong and Y. Ruliang, "SAR target recognition based on MRF and Gabor wavelet feature extraction," in *Proc. IEEE Int. Geosci. Remote Sens. Symp. (IGARSS)*, vol. 2, Jul. 2008, pp. II-907–II-910.

[11] K. Fu, F.-Z. Dou, H.-C. Li, W.-H. Diao, X. Sun, and G.-L. Xu, "Aircraft recognition in SAR images based on scattering structure feature and template matching," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 11, no. 11, pp. 4206–4217, Nov. 2018.

[12] Z. Lin, K. Ji, M. Kang, X. Leng, and H. Zou, "Deep convolutional highway unit network for SAR target classification with limited labeled training data," *IEEE Trans. Geosci. Remote. Sens. Lett.*, vol. 14, no. 7, pp. 1091–1095, Jul. 2017.

[13] D. Malmgren-Hansen, A. Kusk, J. Dall, A. A. Nielsen, R. Engholm, and H. Skriver, "Improving SAR automatic target recognition models with transfer learning from simulated data," *IEEE Geosci. Remote Sens. Lett.*, vol. 14, no. 9, pp. 1484–1488, Sep. 2017.

[14] N. Wang, Y. Wang, H. Liu, Q. Zuo, and J. He, "Feature-fused SAR target discrimination using multiple convolutional neural networks," *IEEE Geosci. Remote Sens. Lett.*, vol. 14, no. 10, pp. 1695–1699, Oct. 2017.

[15] X. Sun, Y. Lv, Z. Wang, and K. Fu, "SCAN: Scattering characteristics analysis network for few-shot aircraft classification in high-resolution SAR images," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5226517.

[16] Y. Sun, Z. Wang, X. Sun, and K. Fu, "SPAN: Strong scattering point aware network for ship detection and classification in large-scale SAR imagery," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 15, pp. 1188–1204, 2022.

[17] P. F. Felzenszwalb, R. B. Girshick, D. McAllester, and D. Ramanan, "Object detection with discriminatively trained part-based models," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 32, no. 9, pp. 1627–1645, Sep. 2010.

[18] Z. Pan, X. Qiu, Z. Huang, and B. Lei, "Airplane recognition in TerraSAR-X images via scatter cluster extraction and reweighted sparse representation," *IEEE Geosci. Remote Sens. Lett.*, vol. 14, no. 1, pp. 112–116, Jan. 2017.

[19] H. Xu, C. Jiang, X. Liang, and Z. Li, "Spatial-aware graph relation network for large-scale object detection," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2019, pp. 9298–9307.

[20] Z. Zhu, X. Sun, W. Diao, K. Chen, G. Xu, and K. Fu, "Invariant structure representation for remote sensing object detection based on graph modeling," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5625217.

[21] C. He, M. Tu, D. Xiong, F. Tu, and M. Liao, "A component-based multi-layer parallel network for airplane detection in SAR imagery," *Remote Sens.*, vol. 10, no. 7, p. 1016, Jul. 2018.

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[22] Y. Hu, Y. Li, and W. Yu, "SAR image recognition based on multi-feature decision fusion for plane," *Mod. Electron. Technique*, vol. 39, pp. 50–55, Jul. 2016.
[23] Z. Pan, N. Jiameng, and Z. Chen, "Recognition method of aircraft target in SAR image based on peak-value matching," *Mod. Electron. Technique*, vol. 38, pp. 19–23, Mar. 2015.
[24] R. Zhang, J. Hong, and F. Ming, "SAR ATR algorithm based on CSAR raw echo modeling," *J. Electron. Inf. Technol.*, vol. 33, no. 1, pp. 27–32, Feb. 2011.
[25] J. Qin, L. Hu, J. Zhou, and Q. Fu, "The profile recognition based on 3D scattering center model," *Radar Sci. Technol.*, vol. 7, pp. 46–51, Jun. 2009.
[26] S. Chen, H. Wang, F. Xu, and Y. Q. Jin, "Target classification using the deep convolutional networks for SAR images," *IEEE Trans. Geosci. Remote Sens.*, vol. 54, no. 8, pp. 4806–4817, Aug. 2016.
[27] S. A. Wagner, "Combination of convolutional feature extraction and support vector machines for radar ATR," in *Proc. 17th Int. Conf. Inf. Fusion (FUSION)*, Jul. 2014, pp. 1–6.
[28] J. Guo, B. Lei, C. Ding, and Y. Zhang, "Synthetic aperture radar image synthesis by using generative adversarial nets," *IEEE Geosci. Remote Sens. Lett.*, vol. 14, no. 7, pp. 1111–1115, Jul. 2017.
[29] Z. Huang, Z. Pan, and B. Lei, "What, where, and how to transfer in SAR target recognition based on deep CNNs," *IEEE Trans. Geosci. Remote Sens.*, vol. 58, no. 4, pp. 2324–2336, Apr. 2020.
[30] K. Fu, T. Zhang, Y. Zhang, Z. Wang, and X. Sun, "Few-shot SAR target classification via metalearning," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 2000314.
[31] M. Gori, G. Monfardini, and F. Scarselli, "A new model for learning in graph domains," in *Proc. IEEE Int. Joint Conf. Neural Netw.*, vol. 2, Jul. 2005, pp. 729–734.
[32] T. N. Kipf and M. Welling, "Semi-supervised classification with graph convolutional networks," 2016, *arXiv:1609.02907*.
[33] P. Veličković, G. Cucurull, A. Casanova, A. Romero, P. Liò, and Y. Bengio, "Graph attention networks," 2017, *arXiv:1710.10903*.
[34] A. Vaswani et al., "Attention is all you need," in *Proc. Adv. Neural Inf. Process. Syst.*, vol. 30, 2017, pp. 1–11.
[35] W. Hamilton, Z. Ying, and J. Leskovec, "Inductive representation learning on large graphs," in *Proc. Adv. Neural Inf. Process. Syst.*, vol. 30, 2017, pp. 1–11.
[36] J. Peng, X. Sun, H. Yu, Y. Tian, C. Deng, and F. Yao, "An instance-based multitask graph network for complex facility recognition in remote sensing imagery," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5615015.
[37] Z. Yuan et al., "MCRN: A multi-source cross-modal retrieval network for remote sensing," *Int. J. Appl. Earth Observ. Geoinf.*, vol. 115, Dec. 2022, Art. no. 103071.
[38] G. Chen, X. Song, H. Zeng, and S. Jiang, "Scene recognition with prototype-agnostic scene layout," *IEEE Trans. Image Process.*, vol. 29, pp. 5877–5888, 2020.
[39] Z.-M. Chen, X.-S. Wei, P. Wang, and Y. Guo, "Multi-label image recognition with graph convolutional networks," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2019, pp. 5177–5186.
[40] Y. Wang et al., "Multi-label classification with label graph superimposing," in *Proc. AAAI Conf. Artif. Intell.*, 2020, vol. 34, no. 7, pp. 12265–12272.
[41] V. Garcia and J. Bruna, "Few-shot learning with graph neural networks," 2017, *arXiv:1711.04043*.
[42] L. Liu, T. Zhou, G. Long, J. Jiang, L. Yao, and C. Zhang, "Prototype propagation networks (PPN) for weakly-supervised few-shot learning on category graph," 2019, *arXiv:1905.04042*.
[43] Z. Yuan et al., "Remote sensing cross-modal text-image retrieval based on global and local information," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5620616.
[44] K. He, X. Zhang, S. Ren, and J. Sun, "Deep residual learning for image recognition," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2016, pp. 770–778.
[45] K. Mikolajczyk and C. Schmid, "Scale & affine invariant interest point detectors," *Int. J. Comput. Vis.*, vol. 60, no. 1, pp. 63–86, 2004.
[46] P. O. Olukanmi, F. Nelwamondo, and T. Marwala, "K-means-MIND: An efficient alternative to repetitive K-means runs," in *Proc. 7th Int. Conf. Soft Comput. Mach. Intell. (ISCMI)*, Nov. 2020, pp. 172–176.
[47] Y. Lv et al., "Scattering information and meta-learning based SAR images interpretation for aircraft target recognition," *J. Radars*, vol. 11, pp. 1–14, Mar. 2022.
[48] X. Li, L. Yu, C.-W. Fu, M. Fang, and P.-A. Heng, "Revisiting metric learning for few-shot image classification," *Neurocomputing*, vol. 406, pp. 49–58, Sep. 2020.
[49] F. Monti, D. Boscaini, J. Masci, E. Rodola, J. Svoboda, and M. M. Bronstein, "Geometric deep learning on graphs and manifolds using mixture model CNNs," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jul. 2017, pp. 5115–5124.
[50] S. Woo, J. Park, J.-Y. Lee, and I. S. Kweon, "CBAM: Convolutional block attention module," in *Proc. Eur. Conf. Comput. Vis. (ECCV)*, 2018, pp. 3–19.
[51] Y. Cao, J. Xu, S. Lin, F. Wei, and H. Hu, "GCNet: Non-local networks meet squeeze-excitation networks and beyond," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. Workshop (ICCVW)*, Oct. 2019, pp. 1971–1980.
[52] N. Ma, X. Zhang, H. Zheng, and J. Sun, "ShuffleNet V2: Practical guidelines for efficient CNN architecture design," in *Proc. Eur. Conf. Comput. Vis. (ECCV)*, 2018, pp. 116–131.
[53] J. Hu, L. Shen, and G. Sun, "Squeeze-and-excitation networks," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, Jun. 2018, pp. 7132–7141.
[54] Y. Wu and K. He, "Group normalization," in *Proc. Eur. Conf. Comput. Vis. (ECCV)*, 2018, pp. 3–19.
[55] J. Dai et al., "Deformable convolutional networks," in *Proc. IEEE Int. Conf. Comput. Vis.*, Oct. 2017, pp. 764–773.
[56] Q.-L. Zhang and Y.-B. Yang, "SA-Net: Shuffle attention for deep convolutional neural networks," in *Proc. IEEE Int. Conf. Acoust., Speech Signal Process. (ICASSP)*, Jun. 2021, pp. 2235–2239.
[57] Y. Wen, K. Zhang, Z. Li, and Y. Qiao, "A discriminative feature learning approach for deep face recognition," in *Proc. Eur. Conf. Comput. Vis.* Berlin, Germany: Springer-Verlag, 2016, pp. 499–515.
[58] K. Fu, J. Fu, Z. Wang, and X. Sun, "Scattering-keypoint-guided network for oriented ship detection in high-resolution and large-scale SAR images," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 14, pp. 11162–11178, 2021.
[59] Y. Kang, Z. Wang, J. Fu, X. Sun, and K. Fu, "SFR-Net: Scattering feature relation network for aircraft detection in complex SAR images," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5218317.
[60] L. Van der Maaten and G. Hinton, "Visualizing data using t-SNE," *J. Mach. Learn. Res.*, vol. 9, no. 11, pp. 2579–2605, 2008.
[61] A. Krizhevsky, I. Sutskever, and G. E. Hinton, "ImageNet classification with deep convolutional neural networks," *Commun. ACM*, vol. 60, no. 6, pp. 84–90, 2017.
[62] K. Simonyan and A. Zisserman, "Very deep convolutional networks for large-scale image recognition," 2014, *arXiv:1409.1556*.
[63] C. Szegedy et al., "Going deeper with convolutions," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2015, pp. 1–9.
[64] C. Szegedy, V. Vanhoucke, S. Ioffe, J. Shlens, and Z. Wojna, "Rethinking the inception architecture for computer vision," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2016, pp. 2818–2826.
[65] M. Tan and Q. Le, "EfficientNet: Rethinking model scaling for convolutional neural networks," in *Proc. Int. Conf. Mach. Learn.*, 2019, pp. 6105–6114.
[66] F. Zhang, M. Li, G. Zhai, and Y. Liu, "Multi-branch and multi-scale attention learning for fine-grained visual categorization," 2020, *arXiv:2003.09150*.
[67] L. Yang, R.-Y. Zhang, L. Li, and X. Xie, "SimAM: A simple, parameter-free attention module for convolutional neural networks," in *Proc. Int. Conf. Mach. Learn.*, 2021, pp. 11863–11874.
[68] J. Ai, Y. Mao, Q. Luo, L. Jia, and M. Xing, "SAR target classification using the multikernel-size feature fusion-based convolutional neural network," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5214313.
[69] M. Zheng et al., "End-to-end object detection with adaptive clustering transformer," 2020, *arXiv:2011.09315*.
[70] N. Carion, F. Massa, G. Synnaeve, N. Usunier, A. Kirillov, and S. Zagoruyko, "End-to-end object detection with transformers," in *Proc. Eur. Conf. Comput. Vis.* Berlin, Germany: Springer-Verlag, 2020, pp. 213–229.



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